Examiner only

SECTION A

Answer **all** questions.

1. Addition of aqueous lead(II) nitrate to aqueous potassium iodide causes a precipitate to form.

(a) Give the colour of the precipitate. [1]

....

(b) Write an ionic equation for the reaction. [1]

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2. Addition of excess hydrochloric acid to aqueous copper(II) sulfate causes the solution to turn green.

Give the formula of the copper-containing species present. [1]

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3. The rate equation for the decomposition of N_2O_5 to form O_2 and NO_2 is first order overall.

(a) Write the rate equation for this reaction. [1]

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(b) Suggest a rate-determining step for this reaction. [1]

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4. State what is meant by the term ‘standard electrode potential’. Include the conditions required. [2]

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5. Phosphorus is able to form two different chlorides, PCl_3 and PCl_5. Nitrogen is only able to form one chloride, NCl_3.

Explain this difference. [1]

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6. Write the expression for the ionic product of water, K_w. [1]

7. Give a reason why the entropy of Hg(l) is greater than that of Au(s). [1]

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